MATHEMATICS

University *of* NORTH ALABAMA

Spring 2026 Math Problem of the Week: Problem #12

Due noon Monday, April 13
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, April 13 or, if you cannot find me, you may leave it with Ms. Phifer in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #12: For fixed positive integer $k \geq 2$, prove that the following infinite series converges to $\ln k$:

$$1 + \frac{1}{2} + \cdots + \frac{1}{k-1} - \frac{k-1}{k} + \frac{1}{k+1} + \cdots$$
$$+ \frac{1}{2k-1} - \frac{k-1}{2k} + \frac{1}{2k+1} + \cdots + \frac{1}{3k-1} - \frac{k-1}{3k} + \frac{1}{3k+1} + \cdots$$